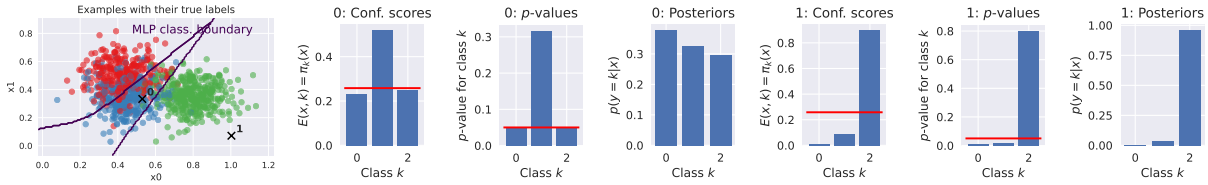

Figure 2: Illustration of CP on a toy dataset detailed in Section 2.2 for two examples X, indexed 0 and 1. Here, the conformity scores $\{E(X,k)\}_{k\in[K]}$ are taken to be the estimated posterior probabilities $\pi_k(X)$ of a multilayer preceptron (MLP) whose decision boundaries are shown. To construct the prediction sets $C(X)$, these scores are thresholded using the threshold τ, cf. Equations (2) and (3). Equivalently, we can use the p-value associated formulation and threshold $(\rho_y)_{y\in[K]}$ at confidence level α, cf Equations (4) and (5). Calibrating against true labels, both approaches obtain coverage $1-\alpha$ (w.r.t. the true labels).

An alternative view on CP can be obtained through a p-value formulation; see e.g. (Sadinle et al., 2019). The conformity scores of the calibration examples and test example (X,Y) are exchangeable so if the distribution of $E(X,Y)$ is continuous then

$$\rho_Y = \frac{|\{i \in [n] : E(X_i,Y_i) \leq E(X,Y)\}| + 1}{n+1} = \frac{\sum_{i=1}^{n} \mathbb{I}[E(X_i,Y_i) \leq E(X,Y)] + 1}{n+1} \tag{4}$$

is uniformly distributed over $\{1/(n+1), 2/(n+1)..., 1\}$ and thus ρ_Y is a p-value in the sense it satisfies $\mathbb{P}(\rho_Y \leq \alpha) \leq \alpha$, equivalently $\mathbb{P}(\rho_Y > \alpha) \geq 1-\alpha$. If the distribution is not continuous, then it can be checked that $\mathbb{P}(\rho_Y \leq \alpha) \leq \alpha$ still holds (see e.g. (Bates et al., 2023)). It follows directly that

$$C(X) := \{y \in [K] : \rho_y > \alpha\} \tag{5}$$

satisfies $\mathbb{P}(Y \in C(X)) \geq 1-\alpha$ and the prediction set obtained this way is identical to the one obtained using Equations (2) and (3). For completeness, the equivalence between both formulations is detailed in Section B.1 of Appendix B. In contrast to calibrating the threshold τ, the p-value formulation requires computing $(\rho_k)_{k\in[K]}$ for each test example X and is thus computationally more expensive. The p-value formulation will be useful in our context as p-values can easily be combined to obtain a new p-value; see e.g. (Vovk & Wang, 2020). We illustrate an application of CP to a 3-class classification problem in Figure 2.

2.2 Motivating Example

We now consider a toy dataset to illustrate the impact a voting strategy can have on prediction sets produced by CP. Consider the true distribution $\mathbb{P} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X} = \mathbb{P}^Y \otimes \mathbb{P}^{X|Y}$ be defined as follows: $\mathbb{P}^{X|Y}$ admits a density $p(x|y) = \mathcal{N}(x; \mu_y, \text{diag}(\sigma_y^2))$ with $\mu_y \in \mathbb{R}^d$ and $\text{diag}(\sigma_y^2) \in \mathbb{R}^{d\times d}$ being mean and variance. Further, one has $\mathbb{P}(Y=y) = w_y$ with $\sum_{y=1}^{K} w_y = 1$. To generate examples $(X,Y) \sim \mathbb{P}$, we first sample Y from $\mathbb{P}(Y=y) = w_y$ and then X from $p(x|y)$ so we have ground truth labels for each example. Furthermore, by Bayes' rule we have access to the true posterior probability mass function $\mathbb{P}(Y=y|X=x)$. If the Gaussians are well-separated, $\mathbb{P}(Y=y|X=x)$ will be crisp, i.e., close to one-hot with low entropy. However, encouraging significant overlap between these Gaussians, e.g., by moving the means $(\mu_k)_{k\in[K]}$ close together, will result in highly ambiguous $\mathbb{P}(Y=y|X=x)$. We sample examples $(X_i,Y_i)_{i\in[n]}$ as outlined above and indicate classes by color in Figure 3 (top left). These synthetic data were previously used in the example displayed in Figure 2 to illustrate CP. The true posterior label probabilities $\mathbb{P}(Y_i=y|X=X_i)$ are also displayed, cf. Figure 3 (top middle). As can be seen, these distributions can be very ambiguous in between all three classes. Let us assume a large set of annotators that allow us by majority vote to recover $\hat{Y}_i = \arg\max_{y\in[K]} \mathbb{P}(Y=y|X=X_i)$, i.e., the voted label, as shown on Figure 3 (top right). This defines the one-hot distribution $\mathbb{P}_{\text{vote}}(Y=y|X=X_i) = \mathbb{I}(y=\hat{Y}_i)$. Clearly, these voted labels ignore the fact that $\mathbb{P}^{Y|X}$ can have high entropy.

Contrary to Figure 2, we now perform standard split CP using calibration data relying on the voted labels $(X_i,\hat{Y}_i)_{i\in[n]}$ from $\mathbb{P}_{\text{vote}} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}_{\text{vote}}$ rather than $(X_i,Y_i)_{i\in[n]}$ from $\mathbb{P} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}$, using a conformity

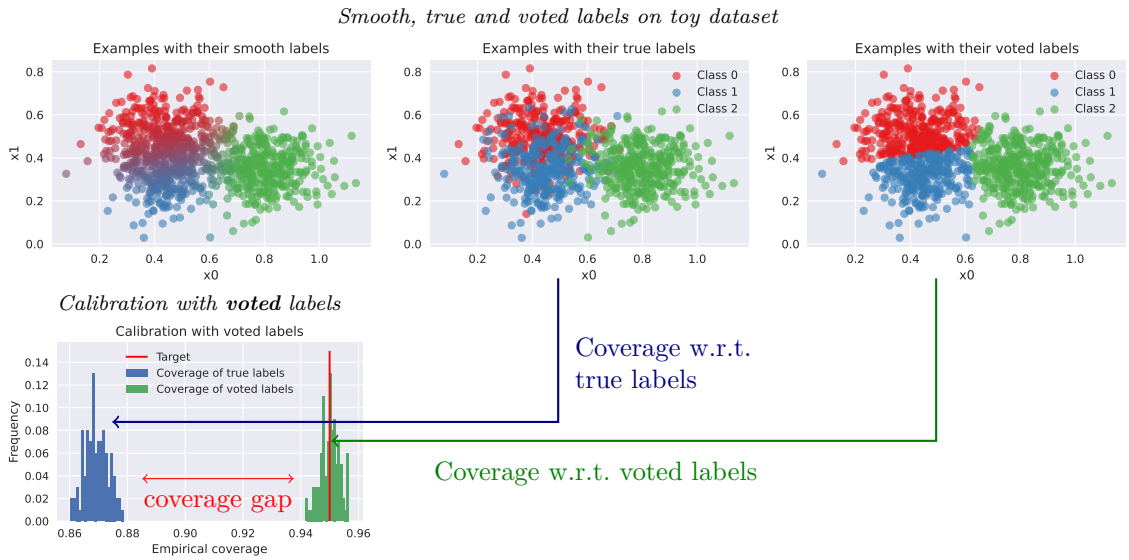

Figure 3: Illustration of an ambiguous problem with $K = 3$ classes, in two dimensions, using our toy dataset. Top: We show examples colored by their true posteriors $\mathbb{P}(Y = y|X = x)$ (left), true labels (middle) and voted labels, i.e., $\hat{Y} = \arg\max_{y\in[K]} \mathbb{P}(Y = y|X = x)$ (right). Note the high ambiguity between the classes, which is clearly ignored in the voted labels. Bottom: Empirical coverage over random calibration/test splits, i.e. the proportion of true or voted labels included in the constructed prediction sets $C(x)$, when calibrating against the voted labels. This produces prediction sets that significantly undercover w.r.t. true labels. Figure 4 shows how our Monte Carlo conformal prediction, as introduced in Section 3, overcomes this gap.

score $E(x, k) = \pi_k(x)$ where $\pi_k(x)$ is given by a multilayer perceptron. We randomly split the examples in two halves for calibration and testing. In Figure 3 (bottom), we plot the empirical coverage, i.e., the fraction of test examples for which (a) the true label (blue) or (b) the voted label (green) is included in the predicted prediction set. In this case, CP guarantees the latter by design, (b), to be 95% (on average across calibration/test splits). Strikingly, however, coverage against the (usually unknown) true labels is significantly worse. Of course, this gap depends on the ambiguity of the problem.

3 Monte Carlo conformal prediction

As discussed in the introduction and illustrated in Section 2.2, we want to avoid using a one-hot distribution $\mathbb{P}^{Y|X}_{\text{vote}}$ when labeling ambiguous examples. Instead we first explain here how, based on expert opinions, we can obtain non-degenerate estimates $\mathbb{P}^{Y|X}_{\text{agg}}$ of $\mathbb{P}^{Y|X}$. We then show how $\mathbb{P}^{Y|X}_{\text{agg}}$ can be leveraged to provide novel CP procedures for constructing prediction sets that satisfy coverage guarantees w.r.t. $\mathbb{P}_{\text{agg}} = \mathbb{P}^X \otimes \mathbb{P}^{Y|X}_{\text{agg}}$. Given that we never observe labels from the true distribution $\mathbb{P}^{Y|X}$, providing guarantees w.r.t. $\mathbb{P}_{\text{agg}}$ is the best we can hope to do. For a test example X, this can be understood as guaranteeing coverage against labels that expert annotators would assign if they had access to X and their annotations were aggregating using $\mathbb{P}^{Y|X}_{\text{agg}}$.

3.1 From expert annotations to $\mathbb{P}^{Y|X}_{\text{agg}}$

From now on, we assume that we have calibration data $(X_i, B_i) \sim \mathbb{P}$ for $i \in [n]$ and a test data $X, B \sim \mathbb{P}$ with B unobserved; the joint distribution of $(X_i, B_i)_{i\in[n]}$ and (X, B) being exchangeable. Here B_i corresponds to a set of expert annotations, the space of annotations being dependent on the application. For example, on CIFAR-10H (Peterson et al., 2019), each expert provides a single label in $[K]$. In contrast, in our dermatology application, using data from (Liu et al., 2020), B_i represents a set of partial rankings whose cardinality depends on i. This corresponds to differential diagnoses from dermatologists. However, while

the format of the B_i's can vary, we are always interested in returning prediction sets of classes $C(X) \subseteq [K]$ satisfying a coverage guarantee w.r.t. $\mathbb{P}_{\text{agg}}$ where $\mathbb{P}_{\text{agg}}$ needs to be specified based on the expert annotations B_i[2]

We give two simple examples to illustrate how $\mathbb{P}_{\text{agg}}^{Y|X}$ can be obtained and then present a more generic framework. However, the aim of this section is not to provide the best way to aggregate expert opinions. This is application dependent and there is a substantial literature on the topic. Instead, we focus here on showing how such techniques can be exploited in a CP framework:

- **Single labels**: We first revisit and extend the construction presented in Section 1 where $B = (Y^1, ..., Y^p)$ with $Y^q \in [K]$ corresponding to the single label the expert q assigns to X. We then consider $\mathbb{P}_{\text{agg}}(Y = y|X, B) = \frac{1}{p}\sum_{k\in[K]} p_k \mathbb{I}(y = k)$ for $p_k = \sum_{q=1}^{p} \mathbb{I}(Y^q = k)$[3]. Note that this approximation is deterministic given (X, B). However, we could also use a bootstrap procedure by resampling the entries of B with replacement.

- **Partial rankings**: In medical applications, it is common for expert annotations to be given by differential diagnoses, corresponding to partial rankings. That is, out of the K possible conditions, each expert returns a partial ranking of conditions because multiple conditions are deemed plausible while a large majority of the conditions can be excluded. While probabilistic models for aggregating such rankings exist (Hajek et al., 2014; Zhu et al., 2023), we follow (Liu et al., 2020) and describe a simple deterministic procedure called inverse rank normalization which we will also exploit in our experiments, see Section 4.1. Let $B = (B^1, ..., B^p)$ be the collection of available partial rankings for data X. Each partial ranking B^q is divided into n_q blocks, $B^q = (B_1^q, ..., B_{n_q}^q)$, with $B_i^q \in\subseteq [K]$ and $B_i^q \bigcap B_j^q = \emptyset$ and $B_{n_q}^q$ the collection of excluded conditions. The conditions in B_i^q are assessed as being more plausible than the conditions in B_j^q for $i < j$. We then define

$$\alpha_y = \sum_{q=1}^{p} \sum_{i=1}^{n_q-1} \frac{1}{i|B_i^q|} \mathbb{I}(y \in B_i^q), \qquad \mathbb{P}_{\text{agg}}(Y = y|X, B) = \frac{\alpha_y}{\sum_{k=1}^{K} \alpha_k}. \tag{6}$$

  As above, this approximation is deterministic given (X, B). Again, we could also use a bootstrapping procedure by resampling the entries of B with replacement.

More generally, we assume the distribution of (X, B) admits a density $p(x, b)$. Then, we will denote by $\lambda = (\lambda_1, ..., \lambda_K)$ the probabilities $(\mathbb{P}_{\text{agg}}(Y = 1|X, B), ..., \mathbb{P}_{\text{agg}}(Y = K|X, B))$ obtained by some deterministic or stochastic annotations aggregation procedures. We refer to λ as *plausibilities* since they quantify how plausible the different classes are to be the actual ground truth label given the annotations B. Then, the resulting marginal density of (X, λ) is given by

$$p(x, \lambda) = \int p(x, \lambda, b)\mathrm{d}b = \int p(\lambda|b, x)p(x, b)\mathrm{d}b. \tag{7}$$

In the examples above, note that $p(\lambda|b, x) = p(\lambda|b)$. Then, our aggregation model for label Y can be written as

$$\mathbb{P}_{\text{agg}}(Y = y|X = x) = \int p(y|\lambda)p(\lambda|x)\mathrm{d}\lambda = \int\int \lambda_y p(\lambda|b, x)p(b|x)\mathrm{d}\lambda\mathrm{d}b. \tag{8}$$

In words, $p(b|x)$ corresponds to the annotation process, i.e., how experts draw their annotations when observing data x, and $p(\lambda|b, x)$ models aggregation of annotations into plausibilities. Given data $(X, B) \sim p(x, b)$, then B is distributed according to $p(b|X)$ given X. So by sampling $\lambda \sim p(\lambda|B, X)$ and then Y such that $\mathbb{P}(Y = y|\lambda) = \lambda_y$, we obtain a sample from $\mathbb{P}_{\text{agg}}(Y = y|X)$. Note however that we cannot obtain i.i.d.

[2]Based on this setup, we could in principle develop a CP method returning prediction sets for B, e.g., prediction sets of rankings, satisfying a coverage guarantee w.r.t $\mathbb{P}$. However, we are interested here in prediction sets for labels.

[3]$\mathbb{P}_{\text{agg}}(Y = y|X, B)$ obviously converges towards $\mathbb{P}(Y = y|X)$ as $p \to \infty$ if $Y^q \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{\text{agg}}(\cdot|X)$, i.e., this assumes that expert annotations follow the true distribution $\mathbb{P}$.